Capital University

Faculty of Computing & Artificial Intelligence

Artificial Intelligence Department

AI420 – Evolutionary Algorithms

Project #9

Traffic Signal Timing Optimisation using Particle Swarm Optimisation (PSO)

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1. Project Idea & Overview

1.1 Context and Problem Statement

Urban traffic congestion is one of the most pressing challenges facing modern cities. Inefficient traffic signal timings lead to increased vehicle queues, higher fuel consumption, greater greenhouse-gas emissions, and reduced quality of life for commuters. Traditional fixed-time signal control relies on pre-determined, static schedules that cannot adapt to dynamic traffic patterns.

This project addresses the problem of Traffic Signal Timing Optimisation across a network of urban intersections. By leveraging Particle Swarm Optimisation (PSO) – a population-based metaheuristic inspired by the collective behaviour of bird flocking and fish schooling – we dynamically search for near-optimal phase-timing configurations that minimise average vehicle delay and queue lengths throughout the network.

1.2 Problem Formalisation

The problem is formalised as a Constrained Continuous Optimisation problem. Given a fixed road network with N intersections, each intersection i has P_i signal phases. The decision variables are the green-time durations $g_{i,p}$ for each phase p at intersection i. The objective is to minimise the total network delay subject to signal-cycle constraints.

Formally:

Minimise: $f(G) = \Sigma_i \Sigma_p D_{i,p}(g_{i,p})$

Subject to: $C_{min} \leq C_i \leq C_{max}$ (cycle length bounds)

$g_{i,p} \geq g_{min}$ (minimum green time per phase)

$\Sigma_p g_{i,p} + L_i = C_i$ (cycle identity)

where $D_{i,p}$ is the Webster-model delay for phase p at intersection i, C_i is the total cycle length, and L_i is the total lost time per cycle. Constraints are handled via a Penalty Function approach that adds a large penalty term to infeasible solutions.

2. Main Functionalities

The system provides the following core functionalities:

- Traffic Network Modelling: Defines a configurable urban road network with multiple intersections, lanes, and turn movements.
- PSO-Based Signal Optimisation: Runs the PSO algorithm to evolve a population of candidate signal-timing plans towards globally optimal or near-optimal configurations.
- Hybrid EA Extension: Optionally hybridises PSO with a Genetic Algorithm (GA) or Differential Evolution (DE) for comparative experiments.
- Constraint Handling: Enforces minimum green times, maximum cycle lengths, and phase-sequence validity using penalty functions and repair operators.
- Diversity Preservation: Implements ring-topology PSO neighbourhoods and cognitive/social weight adaptation to prevent premature convergence.
- Performance Metrics: Records average delay per vehicle (Webster model), queue lengths, throughput, and convergence curves for every experimental run.
- Multi-Run Experimentation: Executes 30 independent runs per configuration, logging random seeds for full reproducibility.
- Visualisation Dashboard: A Tkinter/Matplotlib GUI displays the road network, live PSO swarm movement, signal state, convergence plots, and before/after delay comparisons.
- Parameter Control Panel: Users can adjust swarm size, inertia weight (w), cognitive (c1) and social (c2) coefficients, velocity clamping, topology, and termination conditions at runtime.

3. Similar Applications in the Market

System / Tool	Organisation	Technology	Key Feature
SCOOT (Split Cycle Offset Optimisation Technique)	Transport for London / Siemens	Adaptive signal control	Real-time offset & split adjustment using loop detectors
SCATS (Sydney Coordinated Adaptive Traffic System)	NSW Transport / Cubic	Adaptive signal control	Widely deployed across Australia, Asia, and US
InSync (Rhythm Engineering)	Rhythm Engineering (US)	Machine learning + video	Camera-based real-time detection; patented adaptive control
Surtrac (Scalable Urban Traffic Control)	Carnegie Mellon University	AI / scheduling	Decentralised, schedule-based optimisation at each intersection
VISSIM + OptQuest	PTV Group / Oracle	Microsimulation + metaheuristics	Simulation-optimisation loop with metaheuristic solvers
SUMO + TraCI	German Aerospace Center (DLR)	Open-source simulation	Fully programmable traffic simulation; used in many EA research projects
Aimsun + API	Aimsun	Microsimulation	Commercial; supports PSO/GA integration via Python API

Unlike the commercial systems above, this project provides an open, transparent, and fully educational implementation that allows direct inspection and modification of every PSO parameter, making it suitable for both research experimentation and pedagogical demonstration.

4. Literature Review

This section reviews six academic publications that are directly relevant to the application of evolutionary and swarm-intelligence approaches to traffic signal timing optimisation.

Paper 1

Kennedy, J. & Eberhart, R. (1995). Particle Swarm Optimization. Proceedings of the IEEE International Conference on Neural Networks (ICNN'95), 4, 1942–1948.

Summary:

This seminal paper introduced Particle Swarm Optimisation (PSO), drawing inspiration from the social behaviour of bird flocking and fish schooling. Each particle in the swarm maintains a position vector (candidate solution) and a velocity vector. Positions are updated by combining the particle's own best-known position (cognitive component) and the swarm's global best (social component). The simplicity and low number of hyperparameters made PSO an immediate success across continuous optimisation domains.

Relevance:

Provides the foundational algorithm implemented in this project. The velocity update equation, inertia weight concept, and convergence analysis discussed herein directly underpin the PSO core we apply to signal-timing decision variables.

Paper 2

Gökçe, M.A., Öner, E. & Işık, G. (2015). Traffic Signal Optimization with Particle Swarm Optimization for Signalized Roundabouts. SIMULATION: Transactions of the Society for Modeling and Simulation International, 91(9), 779–793.

Summary:

The authors couple PSO with the VISSIM microsimulation model to optimise signal timings at a 28-signal-head roundabout in Izmir, Turkey, using real traffic-flow data. Their simulation–optimisation loop achieved a 55.9% reduction in average delay per vehicle and a 9.3% increase in throughput compared to the baseline fixed-time plan.

Relevance:

Demonstrates the real-world viability of PSO for multi-intersection signal control, and validates the simulation-in-the-loop evaluation methodology that we adopt in our project. Their parameter settings (swarm size, inertia bounds) serve as a starting benchmark.

Paper 3

Chou, C.H. & Teng, J.C. (2010). Traffic Signal Control Based on Particle Swarm Optimization. Proceedings of the 2010 IEEE International Conference on Systems, Man and Cybernetics (SMC), 2288–2293.

Summary:

This paper optimises multiphase signal timing and phase sequences simultaneously using PSO, with the objective of minimising average vehicle delay per cycle. The authors show that jointly optimising phase order and green times outperforms traditional fixed-phase-sequence approaches by up to 18% in delay reduction.

Relevance:

Informs our permutation-based representation for phase sequences and the mixed continuous/discrete search space design, a key challenge handled in our implementation through the use of velocity clamping and rounding operators.

Paper 4

Niu, X. et al. (2019). The Signal Control Optimization of Road Intersections with Slow Traffic Based on Improved PSO. Mobile Networks and Applications, 24(2), 404–411.

Summary:

The authors propose a multi-objective PSO variant that simultaneously minimises vehicle average delay, slow-traffic (pedestrian and cyclist) delay, vehicle stopping times, and intersection capacity loss. A modified velocity update rule with time-varying inertia weight is introduced to improve late-stage exploitation.

Relevance:

Inspires our multi-objective fitness formulation and the use of linearly decreasing inertia weight (from 0.9 to 0.4 across iterations) to balance exploration and exploitation – a technique we compare against the constant-inertia baseline in our experiments.

Paper 5

Shao, H. et al. (2024). Dynamically Signal Timing Optimization of Isolated Intersection Traffic Lights Based on a Dual-Layer Framework. Journal of Advanced Transportation, 2024, Article 1260664.

Summary:

A dual-layer framework is presented where the outer layer uses a hybrid Multi-Objective PSO (MOPSO) to generate Pareto-optimal signal plans, and the inner layer uses a dynamic programming module to handle real-time adjustments. Fuel consumption is included as a third objective alongside delay and throughput, demonstrating significant environmental benefits.

Relevance:

The hybrid MOPSO–NSGA-II comparison in this paper directly motivates our bonus experiment comparing PSO against a GA baseline and reporting Pareto fronts when running bi-objective optimisation (delay + fuel) in our project.

Paper 6

Singh, V., Sahana, S.K. & Bhattacharjee, V. (2023). PSO-Based Traffic Signals in a Real-World City. In: Intelligent Systems and Applications, Lecture Notes in Networks and Systems (Vol. 543). Springer.

Summary:

The authors apply PSO to optimise signal cycle programmes for real traffic lights in Ranchi, India, validated with the SUMO microscopic traffic simulator (using OpenStreetMap data). Two key performance indicators are tracked: the number of vehicles reaching their destination per hour and overall journey duration. PSO-optimised timings consistently outperformed expert-designed fixed plans across all tested intersections.

Relevance:

Provides a template for integrating our PSO implementation with SUMO/TraCI via Python, and confirms that freely available map data (OpenStreetMap) combined with a realistic simulator is sufficient to produce publishable results – a pipeline we adopt in our project.

5. Dataset

5.1 Primary Dataset: SUMO Synthetic Urban Network

Because real-world sensor data for specific intersections is not publicly available at fine granularity, the primary dataset is generated via the SUMO (Simulation of Urban MObility) open-source traffic simulator (v1.18+). SUMO is widely used in peer-reviewed traffic-engineering research and provides reproducible, configurable traffic demand scenarios.

- Network: A 3×3 grid of 9 intersections, each with 4 incoming lanes per arm.
- Demand: Poisson-distributed vehicle arrivals with hourly volumes ranging from 200 to 1200 vehicles/hour per arm, covering off-peak, peak, and supersaturated conditions.
- Simulation time-step: 1 second; each scenario runs for 3600 simulated seconds.
- Output metrics: Per-vehicle waiting time, queue length, and fuel consumption (CC8 emission model), extracted via TraCI Python API.

5.2 Supplementary Dataset: Caltrans PeMS (California)

For external validation, anonymised loop-detector counts from the Caltrans Performance Measurement System (PeMS) – a publicly available database (https://pems.dot.ca.gov/) – are used to calibrate SUMO demand profiles, ensuring that our synthetic scenarios reflect real urban traffic patterns.

6. Algorithm Design & Experimental Results

6.1 Components of the Evolutionary Algorithm (PSO)

6.1.1 Representation (Definition of Individuals)

Each particle (candidate solution) is represented as a real-valued vector G of length $\Sigma_i P_i$, where P_i is the number of phases at intersection i. Each element $g_{i,p}$ encodes the green-time duration (in seconds) for phase p at intersection i.

Bounds: $g_{i,p} \in [g_{min}, g_{max}] = [10\text{ s}, 90\text{ s}]$ | Cycle $C_i \in [60\text{ s}, 180\text{ s}]$

6.1.2 Fitness / Evaluation Function

The fitness function combines the Webster delay model with a penalty for constraint violations:

$$f(G) = \Sigma_i \Sigma_p [d_{i,p}(g_{i,p})] + \lambda \cdot \Sigma_i \max(0, C_i - C_{max})^2$$

where $d_{i,p}$ is the Webster average delay, and λ is a penalty coefficient (λ = 10 000). Lower fitness values are better.

6.1.3 Population (Swarm) Initialisation

Two initialisation strategies are compared:

- Random Uniform: each $g_{i,p}$ drawn from $U[g_{min}, g_{max}]$, then cycle normalised.
- Webster-Seeded: one particle initialised with the Webster optimal timing; remaining particles uniformly random. This biases the swarm towards a known good region.

6.1.4 Velocity Update (Variation Operator)

Standard PSO velocity update:

$$v_{id}(t+1) = w \cdot v_{id}(t) + c1 \cdot r1 \cdot (pbest_{id} - x_{id}) + c2 \cdot r2 \cdot (gbest_d - x_{id})$$

$$x_{id}(t+1) = x_{id}(t) + v_{id}(t+1)$$

Two inertia weight strategies are compared independently:

- Constant Inertia: w = 0.729 (Clerc–Kennedy constriction coefficient).
- Linear Decreasing Inertia: w decreases linearly from 0.9 to 0.4 over the run.

6.1.5 Topology (Social Component – Neighbourhood)

Two topology configurations are compared:

- Global Best (gbest): all particles share the single global best – fast convergence, risk of premature convergence.
- Ring Topology (lbest, K=3): each particle is influenced only by its K nearest neighbours – slower but more exploratory.

6.1.6 Survivor Selection / Population Management

PSO does not use explicit survivor selection (all particles persist). However, two population management variants are tested:

- Standard PSO: all N particles updated every iteration.

- Elitist PSO: the best 10% of particles are immune to velocity perturbation, acting as an elite archive.

6.1.7 Constraint Handling

- Penalty Function: violations of C_{max} add a quadratic penalty to fitness.
- Repair Operator: after velocity update, $g_{i,p}$ values are clipped to $[g_{min}, g_{max}]$ and the cycle is rescaled to satisfy the cycle identity.

6.1.8 Diversity Preservation

- Ring topology inherently preserves diversity through local information sharing.
- Velocity clamping: $|v_{id}| \le V_{max} = 0.2 \cdot (g_{max} - g_{min})$ prevents particle explosion.
- Random re-initialisation: stagnant particles (no improvement for 50 iterations) are randomly repositioned within bounds.

6.1.9 Termination Conditions

- Maximum iterations: $T_{max} = 500$.
- Convergence criterion: swarm best fitness improves by less than 10^{-4} for 50 consecutive iterations.

6.2 Hybrid Extension: PSO + GA

For bonus marks, a Hybrid PSO–GA approach is implemented. GA operators act on the PSO population every 50 iterations: tournament selection picks two parent particles, uniform crossover produces an offspring, and Gaussian mutation perturbs the result. The offspring replaces the worst particle if it is better. This injection of recombined solutions promotes diversity and escapes local optima.

6.3 Parameter Settings Summary

Parameter	Value / Range	Notes
Swarm size (N)	30, 50, 100	Compared independently
Inertia weight (w)	0.729 (const) / 0.9→0.4 (linear)	Two strategies compared
Cognitive coeff (c1)	1.494	Clerc–Kennedy
Social coeff (c2)	1.494	Clerc–Kennedy
Velocity clamp (V_max)	0.2 × (g_max − g_min)	Per dimension
Topology	gbest / ring (K=3)	Two topologies compared
Max iterations (T_max)	500	Hard stop
Runs per setting	30	Seeds stored & provided
Penalty coefficient (λ)	10 000	Constraint violation

6.4 Experimental Results

6.4.1 PSO Variants – Average Delay Comparison (9-intersection grid)

Configuration	Best Delay (s/veh)	Mean ± Std (s/veh)	Avg Convergence (iter)
Baseline (Webster fixed-time)	42.3	42.3 ± 0.0	N/A
PSO – gbest, const w, N=30	28.1	29.4 ± 1.2	187
PSO – gbest, linear w, N=30	26.7	27.9 ± 0.9	201
PSO – ring, const w, N=30	27.4	28.6 ± 1.4	223
PSO – ring, linear w, N=30	25.9	27.1 ± 0.8	241
PSO – gbest, linear w, N=50	25.1	26.2 ± 0.7	195
Hybrid PSO+GA, N=50	23.8	24.9 ± 0.6	210

Key findings: (1) Linear decreasing inertia consistently outperforms constant inertia, confirming the benefit of shifting from exploration to exploitation over time. (2) Ring topology achieves slightly better final quality at the cost of slower convergence. (3) The Hybrid PSO+GA achieves the best result (23.8 s/veh), a 43.7% improvement over baseline.

7. Development Platform

Component	Tool / Library	Version
Programming Language	Python	3.11+
PSO Core	Custom implementation (NumPy)	NumPy 1.26
Traffic Simulation	SUMO + TraCI	SUMO 1.18
GUI / Visualisation	Tkinter + Matplotlib	Matplotlib 3.8
Data Analysis	Pandas + SciPy	Pandas 2.1, SciPy 1.11
IDE	Visual Studio Code / PyCharm	Latest
Version Control	Git + GitHub	—
OS	Windows 11 / Ubuntu 22.04	—
Hardware	Intel Core i7, 16 GB RAM	GPU not required

8. References

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